

ADVANCED QUANTUM MECHANICS

Fundamentals of quantum mechanics



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DARMSTADT

Claudiu Genes, Institute for Applied Physics; TU Darmstadt, winter semester 2025/2026

LECTURE 2: Heisenberg and interaction pictures. The quantum harmonic oscillator. The particle in a box.

We will review the concept of pictures. In the original Schrödinger picture, the equation gives the evolution of the state vector while operators are frozen. One can instead freeze the state vector and follow the time evolution of operators instead, which is the Heisenberg picture. Alternatively, one can move to a partial picture, where only the free Hamiltonian is removed, i.e. the interaction picture.

Then we address

- the quantum harmonic oscillator in Dirac notation: introducing Fock (number) states, coherent states and thermal mixtures
- the wavefunctions corresponding to the quantum harmonic oscillator's eigenstates
- The particle in a box with periodic boundary conditions and the concept of density of states

From the Schrödinger to the Heisenberg picture. The interaction picture.

We begin with the Schrödinger equation assuming a time independent Hamiltonian, for example something like $\hat{H} = \hat{p}^2/(2m)$. This is different than for example $\hat{H}(t) = \hat{p}^2/(2m) \cos \omega t$, which is an example where explicit time dependence is present and time ordering is needed. Keeping the simple case, we have

$$i\hbar \partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

with $\hat{H}$ time independent. The solution is $|\psi(t)\rangle = U(t) |\psi(0)\rangle$, where the time-evolution operator has the following definition and properties:

$$\hat{U}(t) = e^{-\frac{i}{\hbar} \hat{H} t}, \quad \hat{U}^\dagger(t) \hat{U}(t) = \mathbb{1} \quad \hat{U}^\dagger(t) = \hat{U}^{-1}(t).$$

This shows that we follow the evolution of the state vector, while operators do not evolve in time. There are other options though, where the time evolution is fully or partially shifted to operators.

The Heisenberg picture

In the Heisenberg picture, the state vector is fixed to its value at time $t = 0$

$$|\psi_H\rangle = |\psi(0)\rangle = \hat{U}^{-1}(t) |\psi(t)\rangle.$$

We now ask what happens to operators. We take the operator $\hat{O}(t)$ which may have an explicit time dependence. We ask that expectation values are conserved and use

$$\langle \psi(t) | \hat{O}(t) | \psi(t) \rangle = \langle \psi(0) | U^\dagger(t) \hat{O} U(t) | \psi(0) \rangle = \langle \psi_H | \hat{O}_H(t) | \psi_H \rangle.$$

This indicates the following transformation for any operator which puts all time dependence in the evolution of operators:

$$\hat{O}_H(t) = \hat{U}^\dagger(t) \hat{O}(t) \hat{U}(t).$$

Let us differentiate $\hat{O}_H(t)$ with respect to time:

$$\partial_t \hat{O}_H(t) = (\partial_t \hat{U}^\dagger) \hat{O}(t) \hat{U} + \hat{U}^\dagger (\partial_t \hat{O}(t)) \hat{U} + \hat{U}^\dagger \hat{O}(t) (\partial_t \hat{U}).$$

Using $\partial_t \hat{U} = -\frac{i}{\hbar} \hat{H} \hat{U}$ and $\partial_t \hat{U}^\dagger = \frac{i}{\hbar} \hat{U}^\dagger \hat{H}$, we obtain

$$\begin{aligned} \partial_t \hat{O}_H(t) &= \frac{i}{\hbar} \hat{U}^\dagger \hat{H} \hat{O}(t) \hat{U} + \hat{U}^\dagger (\partial_t \hat{O}(t)) \hat{U} - \frac{i}{\hbar} \hat{U}^\dagger \hat{O}(t) \hat{H} \hat{U} \\ &= \frac{i}{\hbar} [\hat{H}, \hat{O}_H(t)] + (\partial_t \hat{O}(t))_H, \end{aligned}$$

where

$$(\partial_t \hat{O}(t))_H \equiv \hat{U}^\dagger(t) (\partial_t \hat{O}(t)) \hat{U}(t).$$

Heisenberg equations of motion

This gives the general Heisenberg equation of motion:

$$i\hbar \partial_t \hat{O}_H(t) = [\hat{O}_H(t), \hat{H}] + i\hbar (\partial_t \hat{O}(t))_H.$$

If the operator has no explicit time dependence ($\partial_t \hat{O} = 0$), the equation simplifies to

$$i\hbar \partial_t \hat{O}_H(t) = [\hat{O}_H(t), \hat{H}].$$

Let's show this for a simple example where operators have no explicit time dependence, i.e. the case of a free particle. The Hamiltonian is $\hat{H} = \hat{p}^2/2m$ and using the commutation $[\hat{x}, \hat{p}] = i\hbar \mathbb{1}$, we get

$$\partial_t \hat{x}_H = \frac{1}{i\hbar} [\hat{x}_H, \hat{H}] = \frac{\hat{p}_H}{m}, \quad \partial_t \hat{p}_H = \frac{1}{i\hbar} [\hat{p}_H, \hat{H}] = 0.$$

This makes sense as the momentum stays constant

$$\hat{p}_H(t) = \hat{p}_H(0) = \hat{p}, \quad \hat{x}_H(t) = \hat{x}_H(0) + \frac{\hat{p}_H(0)}{m} t = \hat{x} + \frac{\hat{p}}{m} t,$$

while the position operator in the Heisenberg picture gets "infected" with the momentum operator as well.

The interaction picture

Between the Schrödinger and Heisenberg pictures lies the **interaction picture**, particularly useful when the Hamiltonian can be written as a sum of a solvable part and a small perturbation,

$$\hat{H} = \hat{H}_0 + \hat{V}(t).$$

The idea is to remove the trivial, exactly solvable evolution generated by $\hat{H}_0$ and focus on the nontrivial part $\hat{V}(t)$. Operators evolve with $\hat{H}_0$, while states evolve with the interaction term only:

$$\hat{O}_I(t) = e^{\frac{i}{\hbar} \hat{H}_0 t} \hat{O}_S e^{-\frac{i}{\hbar} \hat{H}_0 t}.$$

Interaction picture equations

In the interaction picture the state obeys

$$|\psi_I(t)\rangle = e^{\frac{i}{\hbar}\hat{H}_0 t} |\psi_S(t)\rangle, \quad i\hbar \partial_t |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle,$$

where the transformed Hamiltonian is solely

$$\hat{V}_I(t) = e^{\frac{i}{\hbar}\hat{H}_0 t} \hat{V}(t) e^{-\frac{i}{\hbar}\hat{H}_0 t}.$$

Remember that after solving the Schrödinger equation in the interaction picture, one has to perform the back transformation to retrieve the full evolution of the state vector.

This picture is especially convenient in time-dependent perturbation theory and quantum optics, as it isolates the effect of external driving or coupling while keeping the unperturbed dynamics factored out analytically.

The quantum harmonic oscillator: number (Fock) state basis

Let us now consider a particle of mass m trapped in a harmonic potential of frequency ω , described by the Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2,$$

with the canonical commutation relation

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{1}.$$

Let us now write the general Schrödinger equation for the quantum harmonic oscillator and later ask how the system evolves

$$i\hbar \partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle.$$

As we have mentioned in the previous lecture, the task is to i) diagonalize the Hamiltonian and ii) write the initial state in the eigenstate basis and iii) evolve the overlap coefficients of the initial state with the eigenstates with exponentials containing the eigenvalues of the Hamiltonian.

Starting with the first task one can proceed in more ways. We could write the stationary Schrödinger equation in the position representation

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi(x) = E \psi(x),$$

and then solve a second order differential equation the usual way (see Appendix C for a quick sketch of how that derivation looks like). **Instead, one can use a simpler, more abstract but more elegant solution in terms of a number operator.**

Constructing number (Fock) states - We introduce the following operator and its hermitian conjugate

$$\hat{a} = \frac{1}{\sqrt{2\hbar m\omega}}(m\omega \hat{x} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar m\omega}}(m\omega \hat{x} - i\hat{p}).$$

One can easily check that they satisfy the following commutation relation

$$[\hat{a}, \hat{a}^\dagger] = \mathbb{1},$$

and render the Hamiltonian very simple

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) = \hbar\omega \left(\hat{N} + \frac{1}{2} \right), \quad \hat{N} := \hat{a}^\dagger \hat{a}.$$

The operator $\hat{N}$ is a number operator (see Appendix B for the diagonalization of the number operator). We list here its properties: the eigenvalue problem admits the following solution

$$\hat{N} |n\rangle = n |n\rangle \quad \text{with} \quad n = 0, 1, 2 \dots \infty \quad \text{and} \quad \langle m | n \rangle = \delta_{mn}.$$

The ground state $|0\rangle$ is fixed by $\hat{a}|0\rangle = 0$ (which ensures that no negative eigenvalues of $\hat{N}$ appear). Repeated application of $\hat{a}^\dagger$ generates the entire orthonormal basis:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle, \quad \hat{a}|n\rangle = \sqrt{n} |n-1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1} |n+1\rangle. \quad (1)$$

Energies follow immediately:

$$\hat{H}|n\rangle = E_n|n\rangle, \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \dots \quad (2)$$

Notice that of course the set $\{|n\rangle\}_{n=0}^\infty$ is complete:

$$\sum_{n=0}^\infty |n\rangle\langle n| = \mathbb{1}$$

The quantum harmonic oscillator: Constructing wavefunctions from the number states

We start from the annihilation operator condition

$$\hat{a}|0\rangle = 0, \quad \hat{a} = \frac{1}{\sqrt{2\hbar m\omega}}(m\omega \hat{x} + i\hat{p}).$$

Projecting onto $\langle x|$ and inserting the identity operator written in the eigenbasis of $\hat{x}$,

$$\mathbb{1} = \int_{-\infty}^{\infty} dx' |x'\rangle \langle x'|,$$

we obtain

$$0 = \langle x|\hat{a}|0\rangle = \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{\infty} dx' \langle x|(m\omega \hat{x} + i\hat{p})|x'\rangle \langle x'|0\rangle.$$

Using the position representation of the operators

$$\langle x|\hat{x}|x'\rangle = x' \delta(x - x'), \quad \langle x|\hat{p}|x'\rangle = -i\hbar \partial_{x'} \delta(x - x'),$$

and defining $\psi_0(x') = \langle x'|0\rangle$, the equation becomes

$$0 = \frac{1}{\sqrt{2\hbar m\omega}} \int dx' \left[m\omega x' \delta(x - x') - \hbar \partial_{x'} \delta(x - x') \right] \psi_0(x').$$

Using the distribution properties

$$\int dx' x' \delta(x - x') \psi_0(x') = x \psi_0(x), \quad \int dx' \partial_{x'} \delta(x - x') \psi_0(x') = -\partial_x \psi_0(x),$$

we obtain the differential equation

$$(m\omega x + \hbar \partial_x) \psi_0(x) = 0.$$

Its solution is

$$\psi_0(x) = A \exp\left(-\frac{m\omega}{2\hbar} x^2\right),$$

where A is a normalization constant. Enforcing

$$\int_{-\infty}^{\infty} |\psi_0(x)|^2 dx = 1$$

gives

$$A = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}.$$

Thus the normalized ground-state wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right).$$

Let's continue with constructing the next wavefunction of the first level

$$\psi_1(x) = \langle x|1\rangle = \langle x|\hat{a}^\dagger|0\rangle.$$

We now insert the identity operator written in the eigenbasis of $\hat{x}$,

$$\mathbb{1} = \int_{-\infty}^{\infty} dx' |x'\rangle \langle x'|,$$

to obtain

$$\psi_1(x) = \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{\infty} dx' \langle x|(m\omega\hat{x} - i\hat{p})|x'\rangle \langle x'|0\rangle.$$

Proceeding as before by replacing the matrix elements of the position and momentum operators and integrating over the delta distributions, we get to obtain

$$\psi_1(x) = \frac{1}{\sqrt{2\hbar m\omega}} [m\omega x - \hbar\partial_x] \psi_0(x) = \sqrt{\frac{2m\omega}{\hbar}} x \psi_0(x).$$

We see that it is easy to obtain the whole spectrum of the Hamiltonian in terms of wavefunctions by repeated operation on the previous states. We go one step further starting from

$$|2\rangle = \frac{1}{\sqrt{2}} \hat{a}^\dagger |1\rangle, \quad \psi_2(x) = \langle x|2\rangle = \frac{1}{\sqrt{2}} \langle x|\hat{a}^\dagger|1\rangle.$$

and following the same steps to get

$$\psi_2(x) = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{\infty} dx' \langle x|(m\omega\hat{x} - i\hat{p})|x'\rangle \langle x'|1\rangle.$$

and then evaluate that

$$\psi_2(x) = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2\hbar m\omega}} [m\omega x - \hbar\partial_x] \psi_1(x).$$

This can be rewritten as a recurrence connecting to the first and zero-th order wavefunctions:

$$\psi_2(x) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{2m\omega}{\hbar}} x \psi_1(x) - \psi_0(x) \right).$$

General wavefunction solutions of the quantum harmonic oscillator

We can now state the general recurrence equation

$$\psi_n(x) = \frac{1}{\sqrt{2n}} \left(\sqrt{\frac{2m\omega}{\hbar}} x \psi_{n-1}(x) - \sqrt{2(n-1)} \psi_{n-2}(x) \right).$$

This is connected to the standard Hermite recurrence relation, and the wavefunctions can be written as

$$\psi_n(x) = \frac{1}{\pi^{1/4} \sqrt{2^n n!}} \left(\frac{m\omega}{\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) \exp\left(-\frac{m\omega x^2}{2\hbar}\right)$$

in terms of well known (and widely listed and in numerical programs implemented) [Hermite polynomials](#).

The easiest way to understand the wavefunctions of the quantum harmonic oscillator is to visualize them (which is easily accomplished by using [available online resources](#)).

A few interesting quantum states of the quantum harmonic oscillator

One can drive a quantum mechanical oscillator into a variety of interesting states

- Fock or number states $|n\rangle$: interesting as so-called non-classical states.
- coherent (or so-called classical) states: useful to describe laser light
- squeezed states: useful in metrology for optimizing signal to noise ratio
- thermal states: the natural state of an oscillator in contact with a thermal bath

We will only describe here two situations: coherent states ([pure superpositions](#) of state vectors) and thermal states ([statistical mixtures](#) - described by density operators instead of state vectors).

Coherent state

A coherent state is defined as a right eigenstate of the annihilation operator (as the $\hat{a}$ operator is non-Hermitian, it has different [left and right eigenvectors](#)):

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \alpha \in \mathbb{C}.$$

These states can be obtained by applying the displacement operator $\hat{D}(\alpha)$ to the ground state:

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle, \quad \hat{D}(\alpha) = \exp(\alpha\hat{a}^\dagger - \alpha^*\hat{a}).$$

Expanding in the number-state basis gives

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

These states are sometimes described as classical: at large amplitude, they approximately describe the motion of a classical harmonic oscillator. We won't use them much but you will encounter them extensively in the field of quantum optics, as they provide a good description of quasi-monochromatic laser light.

Thermal state

A thermal state describes a quantum system in thermal equilibrium (established by putting it in contact with some very large bath) at temperature T . It is defined by the density operator

$$\hat{\rho}_{\text{th}} = \frac{e^{-\beta\hat{H}}}{Z} = \frac{1}{Z} \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+1/2)} |n\rangle\langle n|, \quad \beta = \frac{1}{k_B T},$$

where $Z = \text{Tr}(e^{-\beta\hat{H}})$ is the partition function. We have used the fact that the Hamiltonian is diagonal in the number-state basis,

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) = \sum_{n=0}^{\infty} \hbar\omega\left(n + \frac{1}{2}\right) |n\rangle\langle n|.$$

The partition function, obtained as the trace over all diagonal elements, is

$$Z = \sum_{n=0}^{\infty} \langle n| e^{-\beta\hat{H}} |n\rangle = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+1/2)} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}.$$

Hence, the thermal density operator is diagonal in the number basis:

$$\hat{\rho}_{\text{th}} = \sum_{n=0}^{\infty} \frac{\bar{n}^n}{(1 + \bar{n})^{n+1}} |n\rangle\langle n|,$$

where the mean occupation number (proof left for Exercise 3) is

$$\bar{n} = \frac{1}{e^{\beta\hbar\omega} - 1}.$$

Thus, a thermal state is an **incoherent statistical mixture of number states**, each weighted by the Boltzmann factor $e^{-\beta E_n}$, and describes the probability distribution over energy levels of a **single particle** described as a quantum harmonic oscillator in equilibrium. There is a subtle difference to the Bose–Einstein distribution, which has the same formula but describes the **population distribution of a many body system of bosons** over trap states.

The free particle in a box with periodic boundary conditions

We consider a single particle of mass m confined between $-L/2$ and $+L/2$.

We start from the definition of the momentum eigenkets,

$$\hat{p} |k\rangle = \hbar k |k\rangle,$$

where the wavevector k takes a discrete set of values determined by the boundary conditions.

Projecting onto the position basis,

$$\langle x | \hat{p} | k \rangle = \hbar k \langle x | k \rangle, \quad \phi_k(x) \equiv \langle x | k \rangle.$$

Inserting the identity operator written in the position basis,

$$\mathbb{1} = \int_{-L/2}^{L/2} dx' |x'\rangle \langle x'|,$$

we can write

$$\phi_k(x) = \frac{1}{\hbar k} \langle x | \hat{p} | k \rangle = \frac{1}{\hbar k} \int_{-L/2}^{L/2} dx' \langle x | \hat{p} | x' \rangle \langle x' | k \rangle = \frac{-i}{k} \int_{-L/2}^{L/2} dx' (\partial_x \delta(x - x')) \phi_k(x') = \frac{-i}{k} \partial_x \phi_k(x).$$

The general solution is

$$\phi_k(x) = C_k e^{ikx}.$$

Introducing boundary conditions - Boundary conditions are important. One has two choices here:

- **Periodic boundary conditions (PBC):** simulate an infinite, translationally invariant system; useful for modeling bulk materials or solids. We will use it later to simplify the description of a free particle in 3D by making our life easy and considering discrete eigenstates of the momentum operator (when we tackle scattering theory).
- **Open boundary conditions (OBC):** describe a finite system with physical walls; useful for isolated particles or quantum wells, gives rise to sine and cosine eigenstates.

Periodic boundary conditions $\phi_k(x + L) = \phi_k(x)$ impose

$$e^{ikL} = 1 \quad \Rightarrow \quad k = \frac{2\pi}{L}n, \quad n \in \mathbb{Z}.$$

Normalization over the interval $[-L/2, L/2]$ requires

$$\int_{-L/2}^{L/2} dx \phi_k^*(x) \phi_{k'}(x) = \delta_{k,k'} \quad \Rightarrow \quad C_k = \frac{1}{\sqrt{L}}.$$

General wavefunction solutions of the 1D free particle in a box

The wavefunctions of the free particle in a box with periodic boundary conditions are given by

$$\phi_k(x) = \langle x | k \rangle = \frac{1}{\sqrt{L}} e^{ikx}, \quad k = \frac{2\pi}{L}n, \quad n \in \mathbb{Z}.$$

Let us also remark on the density of states in k -space. If one looks at the distance between two consecutive allowed k values, this is simply $2\pi/L$. This means that there is one state within this interval. Hence, the density of states reads

$$g^{1D}(k) \equiv \frac{dN}{dk} = \frac{1}{2\pi/L} = \frac{L}{2\pi}.$$

The generalization to three dimensions is straightforward, as the three directions are independent. The density of states will then correspond to one single available state divided by the volume of a cube of side $2\pi/L$, giving

$$g^{3D}(\mathbf{k}) \equiv \frac{dN}{d\mathbf{k}} = \frac{1}{(2\pi/L)^3} = \left(\frac{L}{2\pi}\right)^3 = \frac{V}{(2\pi)^3}$$

General wavefunction solutions of the 3D free particle in a box

The wavefunctions of a free particle in a three-dimensional box with periodic boundary conditions are given by

$$\phi_{\mathbf{k}}(\mathbf{r}) = \langle \mathbf{r} | \mathbf{k} \rangle = \frac{1}{\sqrt{V}} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad \mathbf{k} = \frac{2\pi}{L} (n_x, n_y, n_z), \quad n_x, n_y, n_z \in \mathbb{Z},$$

where $V = L^3$ is the volume of the box. Each allowed $\mathbf{k}$ corresponds to a distinct plane-wave mode satisfying periodic boundary conditions in all three spatial directions.

A. Appendix: Heisenberg picture for a two-level system

We consider a two-level system described by the Hamiltonian

$$\hat{H} = \frac{\hbar\omega}{2} \sigma_z, \quad \text{with} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The time-evolution operator is (this is easy to compute as the σ_z is diagonal so its exponential is also diagonal)

$$\hat{U}(t) = e^{-\frac{i}{\hbar} \hat{H} t} = e^{-i \frac{\omega t}{2} \sigma_z} = \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{+i\omega t/2} \end{pmatrix}.$$

In the Schrödinger picture, the operator σ_x is time independent:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

In the Heisenberg picture, it evolves as

$$\sigma_x^{(H)}(t) = \hat{U}^\dagger(t) \sigma_x \hat{U}(t) = \begin{pmatrix} 0 & e^{+i\omega t} \\ e^{-i\omega t} & 0 \end{pmatrix}.$$

Let the initial state at $t = 0$ be

$$|\psi(0)\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Schrödinger picture. The state evolves as

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle = \begin{pmatrix} \alpha e^{-i\omega t/2} \\ \beta e^{+i\omega t/2} \end{pmatrix}.$$

The operator is fixed, so

$$\langle \sigma_x \rangle(t) = \langle \psi(t) | \sigma_x | \psi(t) \rangle = \alpha^* \beta e^{+i\omega t} + \beta^* \alpha e^{-i\omega t}.$$

Heisenberg picture. The state is fixed, and the operator evolves:

$$\langle \sigma_x \rangle_H(t) = \langle \psi(0) | \sigma_x^{(H)}(t) | \psi(0) \rangle = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix} \begin{pmatrix} e^{+i\omega t} & \beta \\ e^{-i\omega t} & \alpha \end{pmatrix} = \alpha^* \beta e^{+i\omega t} + \beta^* \alpha e^{-i\omega t}.$$

Both pictures yield exactly the same physical result confirming that expectation values of observables are invariant under the choice of picture.

B. Appendix: Finding eigenvalues and eigenstates for the number operator

The operator

$$\hat{N} = \hat{a}^\dagger \hat{a}$$

is Hermitian and has non-negative eigenvalues, since for any state $|\psi\rangle$,

$$\langle\psi|\hat{N}|\psi\rangle = \langle\psi|\hat{a}^\dagger\hat{a}|\psi\rangle = \|\hat{a}|\psi\rangle\|^2 \geq 0.$$

Because $\hat{N}$ is Hermitian, it can be diagonalized and its eigenstates form a complete orthonormal basis. We define these eigenstates by

$$\hat{N}|n\rangle = n|n\rangle, \quad n \geq 0.$$

From the commutation relations

$$[\hat{N}, \hat{a}] = -\hat{a}, \quad [\hat{N}, \hat{a}^\dagger] = +\hat{a}^\dagger,$$

it follows that

$$\hat{N}(\hat{a}|n\rangle) = (n-1)(\hat{a}|n\rangle), \quad \hat{N}(\hat{a}^\dagger|n\rangle) = (n+1)(\hat{a}^\dagger|n\rangle),$$

which shows that $\hat{a}$ lowers and $\hat{a}^\dagger$ raises the eigenvalue of $\hat{N}$ by one unit.

Because the spectrum of $\hat{N}$ must remain non-negative, there exists a lowest state $|0\rangle$ such that

$$\hat{a}|0\rangle = 0.$$

All higher eigenstates can be generated by repeated application of the creation operator,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \dots,$$

and satisfy

$$\hat{N}|n\rangle = n|n\rangle.$$

The Hamiltonian is diagonal in this basis,

$$\hat{H}|n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle,$$

demonstrating that $\hat{N}$ is indeed the number operator counting the excitation quanta of the harmonic oscillator.

C. Appendix: Short derivation of the harmonic oscillator eigenvalues and eigenstates

We start from the stationary Schrödinger equation

$$-\frac{\hbar^2}{2m} \psi''(x) + \frac{1}{2} m \omega^2 x^2 \psi(x) = E \psi(x).$$

Introduce the oscillator length

$$\ell = \sqrt{\frac{\hbar}{m\omega}}, \quad \xi = \frac{x}{\ell},$$

so that $\psi''(x) = \ell^{-2} \psi''(\xi)$. The equation becomes

$$\left[-\frac{1}{2} \frac{d^2}{d\xi^2} + \frac{1}{2} \xi^2 \right] \psi(\xi) = \varepsilon \psi(\xi), \quad \varepsilon = \frac{E}{\hbar\omega}.$$

For $|\xi| \rightarrow \infty$, the potential term dominates, and normalizable solutions must decay as $e^{-\xi^2/2}$. We therefore set

$$\psi(\xi) = e^{-\xi^2/2} u(\xi).$$

Substituting this ansatz yields

$$u''(\xi) - 2\xi u'(\xi) + 2\left(\varepsilon - \frac{1}{2}\right) u(\xi) = 0,$$

which is the Hermite differential equation.

We expand $u(\xi)$ as a power series,

$$u(\xi) = \sum_{k=0}^{\infty} c_k \xi^k,$$

and obtain the recurrence relation

$$c_{k+2} = \frac{2(k + \frac{1}{2} - \varepsilon)}{(k+1)(k+2)} c_k.$$

For the series to terminate (ensuring normalizability), the numerator must vanish after a finite number of steps:

$$\varepsilon - \frac{1}{2} = n \quad \Rightarrow \quad \varepsilon = n + \frac{1}{2}, \quad n = 0, 1, 2, \dots$$

In this case, $u(\xi)$ becomes a polynomial of degree n , the Hermite polynomial $H_n(\xi)$.

Thus the normalized eigenfunctions and energies are

$$\psi_n(x) = \frac{1}{\pi^{1/4} \sqrt{2^n n!} \ell} \exp\left(-\frac{x^2}{2\ell^2}\right) H_n\left(\frac{x}{\ell}\right), \quad E_n = \hbar\omega \left(n + \frac{1}{2}\right).$$

These eigenfunctions form an orthonormal basis of stationary states for the harmonic oscillator.